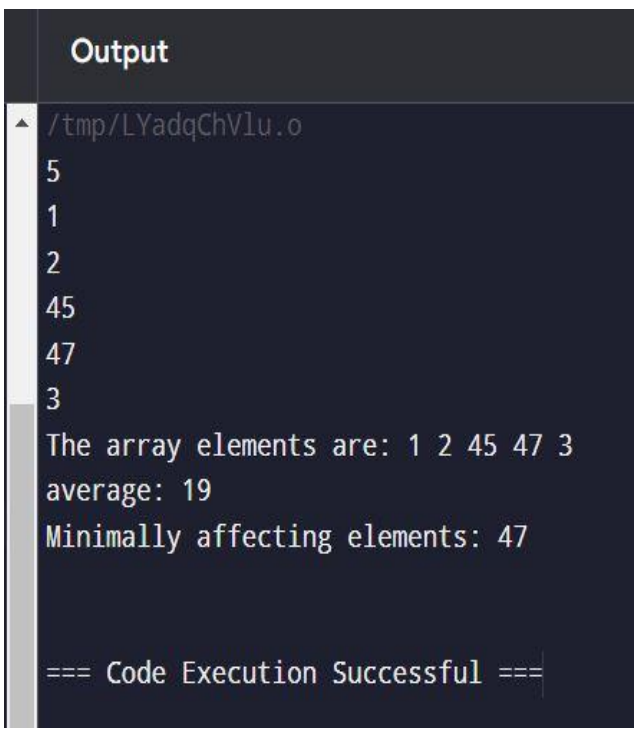
C LEVEL 3 - ARRAY

MINIMALLY AFFECTING NUMBER:

```
#include <stdio.h>
#include<stdlib.h>

int main() {
    int size, sum = 0, average;
    scanf("%d", &size);
    int arr[size];
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }
    average = sum / size;

    printf("The array elements are: ");
    for (int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    printf("average: %d\n", average);

    int max_diff = 0;
    int minimally_affecting_elements[size];
    int count = 0;

    for (int i = 0; i < size; i++) {
        int diff = abs(arr[i] - average);
        if (diff > max_diff) {
            max_diff = diff;
        }
    }

    for (int i = 0; i < size; i++) {
```

```
        int diff = abs(arr[i] - average);
        if (diff == max_diff) {
            minimally_affecting_elements[count] = arr[i];
            count++;
        }
    }

    printf("Minimally affecting elements: ");
    for (int i = 0; i < count; i++) {
        printf("%d ", minimally_affecting_elements[i]);
    }
    printf("\n");

    return 0;
}
```



DUPLICATE NUMBER COUNT:

```
#include <stdio.h>

int main() {
```

```
    int arr[100]; // Assuming the maximum size of the array is 100
    int size;
    printf("Enter the array size: ");
    scanf("%d", &size);
    if (size <= 0) {
        printf("Invalid input: Array size should be greater than 0.\n");
        return 1;
    }
    printf("Enter the array elements: ");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    int duplicateCount = 0;
    for (int i = 0; i < size; i++) {
        for (int j = i + 1; j < size; j++) {
            if (arr[i] == arr[j]) {
                duplicateCount++;
                break;
            }
        }
    }
    printf("Duplicate count = %d\n", duplicateCount);

    return 0;
}
```

```
/tmp/P0fdf9w6HB.o
Enter the array size: 5
Enter the array elements: -1
-1
-2
-2
-2
Duplicate count = 3


=== Code Execution Successful ===
```

EVEN ELEMENTS

```
#include<stdio.h>
int main(){
    int size,count=0;
    scanf("%d",&size);
    int arr[size];
    for(int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
     for(int i=0;i<size;i++){
         if(arr[i]%2==0){
            count++;
         }
     }
     int even_arr[count];
    int j=0;
    for(int i=0;i<size;i++){
         if(arr[i]%2==0){
            even_arr[j]=arr[i];
            j++;
         }
    }
     for(int i=0;i<count;i++){
         printf("%d ",even_arr[i]);
     }
     return 0;
}
```

```
/tmp/72FEITIcOU.o
5
1
2
45
44
65
2 44

=== Code Execution Successful ===
```

FREQUENCY COUNTER:

```
#include <stdio.h>

int main() {
    int A[100]; // Assuming the maximum size of the array is 100
    int n;
    printf("Enter the array size: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid input: Array size should be greater than 0.\n");
        return 1;
    }

    printf("Enter the array elements: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int frequency[100] = {0}; // Initialize frequency array with zeros
    for (int i = 0; i < n; i++) {
        int found = 0;
        for (int j =0; j < i; j++) {
            if (A[i] == A[j]) {
                found = 1;
                break;
            }
        }
        if (!found) {
            frequency[A[i]]++; // Increment frequency of unique element
        } else {
            frequency[A[i - 1]]++; // Increment frequency of duplicate element
        }
    }

    printf("Frequency of each element:\n");
    for (int i = 0; i < 100; i++) {
```

```
        if (frequency[i] > 0) {
            printf("Element %d: Frequency = %d\n", i, frequency[i]);
        }
    }

    return 0;
}
```

```
/tmp/TaoxLWAvYw.o
Enter the array size: 5
Enter the array elements: 1
1
2
2
3
Frequency of each element:
Element 1: Frequency = 2
Element 2: Frequency = 2
Element 3: Frequency = 1


=== Code Execution Successful ===
```

GREATER THAN AVG:

```
#include<stdio.h>
int main(){
    int size,sum=0,avg,count=0;
    scanf("%d",&size);
    int arr[size];
    for(int i=0;i<size;i++){
        scanf("%d",&arr[i]);
        sum+=arr[i];
    }
    avg=sum/size;
    printf("average is: %d\n",avg);
    for(int i=0;i<size;i++){
        if(avg<arr[i]){
            count++;
        }
    }
    int arr_2[count];
```

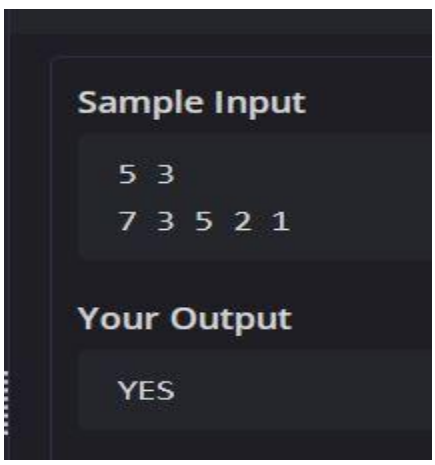
```
    int j=0;
    for(int i=0;i<size;i++){
        if(avg<arr[i]){
            arr_2[j]=arr[i];
            j++;
        }

    }
     printf("elements which are greater than the average:");
    for(int i=0;i<count;i++){
        printf(" %d",arr_2[i]);
    }
    return 0;

}
```

```
/tmp/bt9fs8CkrX.o
5
1
2
3
4
5
average is: 3
elements which are greater than the average: 4 5

=== Code Execution Successful ===
```

SEARCH AN ELEMENT IN ARRAY:

```
#include <stdio.h>
int main() {
        int size,search;
        int count=0;
        scanf("%d%d",&size,&search);
        int arr[size];
        for (int i=0;i<size;i++){
```

```
        scanf("%d",&arr[i]);

        if(search==arr[i]){
            count=1;
            break;
        }
    }
    if(count==1){
        printf("YES");
    }else{
        printf("NO");
    }
    return 0;

}
```



REVERSE ARRAY:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int size;
    scanf("%d",&size);
    int arr[size];
```

```
    for(int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=size-1;i>=0;i--){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

Input (stdin)

```
6
16 13 7 2 1 12
```

Your Output (stdout)

```
12 1 2 7 13 16
```

Expected Output

```
12 1 2 7 13 16
```

ADDITION OF 2 ARRAY:

```
#include<stdio.h>
int main(){
   int size1,size2,size3;
   scanf("%d%d",&size1,&size2);

   if(size1>size2){
      size3=size1;
   }else{
      size3=size2;
   }
   int arr1[size1],arr2[size2],arr3[size3];
```

```
    printf("elements of array1:\n");
    for(int i=0;i<size1;i++){
        scanf("%d",&arr1[i]);
    }
      printf("elements of array2:\n");
  for(int i=0;i<size2;i++){
        scanf("%d",&arr2[i]);
    }
    printf("array 1:");
    for(int i=0;i<size1;i++){
        printf("%d ",arr1[i]);
    }
    printf("\narray 2: ");
    for(int i=0;i<size2;i++){
        printf("%d ",arr2[i]);
    }
        for(int i = 0; i < size3; i++){
    if(i < size1 && i < size2){
        arr3[i] = arr1[i] + arr2[i];
    }else if(i < size1){
        arr3[i] = arr1[i];
    }else{
        arr3[i] = arr2[i - size1];
    }
}
printf("\narray 3: ");
for(int i=0;i<size3;i++){
       printf("%d ",arr3[i]);
    }
    return 0;
}
```

```
/tmp/Iflmm2OPt6.o
5
4
elements of array1:
1
2
3
4
5
elements of array2:
1
2
3
4
array 1:1 2 3 4 5
array 2: 1 2 3 4
array 3: 2 4 6 8 5

=== Code Execution Successful ===
```

MAXIMUM IN ARRAY:

```
#include<stdio.h>
int main(){
    int t,n;

    scanf("%d",&t);
    for(int i=0;i<t;i++){

        scanf("%d",&n);
        int arr[n];


        for(int j=0;j<n;j++){
            scanf("%d",&arr[j]);
        }
        int max=arr[0];
        for(int j=0;j<n;j++){
            if(arr[j]>max){
                max=arr[j];
```

```
        }
    }


  printf("%d\n",max);
 }

 return 0;
}
```

Sample Input

```
1
5
4 7 6 3 1
```

Your Output

```
7
```

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases. The description of the test cases follows.
- Each test case consists of two lines of input.
  - The first line of the test case contains three space-separated integers — N, X, and Y.
  - The second line contains N space-separated integers — $A_1, A_2, \ldots, A_N$.

Output Format

- For each test case, output COUPON if Chef should buy the discount coupon, and NO COUPON otherwise.

Each letter of the output may be printed in either lowercase or uppercase. For example, the strings coupon, CouPoN, and COUPON will all be treated as equivalent.

Constraints

- $1 \leq T \leq 1000$
- $1 \leq N \leq 100$
- $1 \leq X, Y \leq 10^5$
- $1 \leq A_i \leq 10^5$

```
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int n, x, y;
        scanf("%d %d %d", &n, &x, &y);
        int a[n];

        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }
        int sum=0,sum2=0,sum3=0;
         for (int i = 0; i < n; i++) {
           sum+=a[i];
           a[i]=a[i]-y;
           if(a[i]<0){
               a[i]=0;
           }
           sum2+=a[i];
          sum3=sum2+x;
        }

        if(sum<=sum3){
            printf("\nNO COUPON");

        }else{
            printf("\nCOUPON");

        }
    }

    return 0;
}
```

Input	Output
5 4 30 10 15 8 22 6 4 40 10 15 8 22 6 4 34 10 15 8 22 6 2 10 100 60 80 3 30 5	COUPON NO COUPON NO COUPON COUPON NO COUPON

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.
  - The first line of each test case contains two space-separated integers N and X — the number of items and the minimum freshness value an item should have.
  - The second line contains N space-separated integers, the array A, denoting the freshness value of each item.
  - The third line contains N space-separated integers, the array B, denoting the cost of each item.

Output Format

For each test case, output on a new line, the total cost of the groceries Chef buys.

Constraints

- $1 \leq T \leq 100$
- $1 \leq N, X \leq 100$
- $1 \leq A_i, B_i \leq 100$

```
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int n, x;
        scanf("%d %d", &n, &x);
        int a[n], b[n];

        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }

        for (int j = 0; j < n; j++) {
            scanf("%d", &b[j]);
        }
         int sum=0;
        for (int i = 0; i< n; i++) {
            if(a[i]>=x){
                sum+=b[i];
            }

        }
         printf("%d ",sum);
    }
    return 0;
}
```

Input	Output
4 2 20 15 67 10 90 3 1 1 2 3 1 2 3 3 100 10 90 50 30 7 93 4 50 12 78 50 40 40 30 20 10	90 6 0 50

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.
  - The first line of each test case contains a single integer N - the size of the array.
  - The next line of each test case contains N space-separated integers $A_1, A_2, \ldots, A_N$ - the elements of the array.

Output Format

For each test case, output on a new line, the **minimum** number of operations required to make M the **maximum** value in the array A.

Constraints

- $1 \leq T \leq 100$
- $1 \leq N \leq 100$
- $1 \leq A_i \leq 100$

```
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int n;
        scanf("%d", &n);
        int a[n];

        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }

        int min=a[0];
```

```
        for (int i = 0; i < n; i++) {

            if(a[i]<min){
                min=a[i];
            }
        }

        int count = 0;
        for (int i = 0; i < n; i++) {
            if (a[i] != min) {
                count++;
            }
        }
printf("\n%d",count);
}


    return 0;
}
```

Input	Output
3 2 1 2 4 2 2 3 4 1 1	1 2 0